GLOBAL ECONOMICS

10th grade

Term 2 – Lesson 3 MATERIAL: Decision Analysis with Bayes, Maximin, and Minimax Regret (Worked Solutions)

C10: Full end-to-end decision analysis

1.

Problem description

A food-processing company must choose one packaging strategy for next quarter: P (premium line), Q (standard line), or R (hybrid line). Demand can be high (0.60) or low (0.40). Operating cost has two states: stable input costs (0.65) and costly input spike (0.35). Revenues (thousand USD):

- High demand: $P = 500,\ Q = 540,\ R = 520.$
- Low demand: $P = 320,\ Q = 300,\ R = 310.$

Costs (thousand USD):

- P: stable 190, spike 255.
- Q: stable 220, spike 290.
- R: stable 205, spike 270.

The stable-cost values are lower because packaging film contracts lock in most input prices, while the spike state captures imported resin and freight increases. Build the payoff table, apply EV, Maximin, and Minimax Regret, produce ordered rankings, then use a staged decision tree to choose a final strategy.

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Step 1. Build the payoff table from narrative data. Joint state probabilities:

$$\Pr(H, S) = 0.60 \cdot 0.65 = 0.39, \quad \Pr(H, C) = 0.60 \cdot 0.35 = 0.21,$$
$$\Pr(L, S) = 0.40 \cdot 0.65 = 0.26, \quad \Pr(L, C) = 0.40 \cdot 0.35 = 0.14.$$

Payoff = revenue − cost.

State of nature	Probability	P	Q	R
High demand, stable cost	0.39	$500 - 190 = 310$	$540 - 220 = 320$	$520 - 205 = 315$
High demand, costly spike	0.21	$500 - 255 = 245$	$540 - 290 = 250$	$520 - 270 = 250$
Low demand, stable cost	0.26	$320 - 190 = 130$	$300 - 220 = 80$	$310 - 205 = 105$
Low demand, costly spike	0.14	$320 - 255 = 65$	$300 - 290 = 10$	$310 - 270 = 40$

Step 2. Bayes criterion (EV).

$$EV(P) = 208.95, \quad EV(Q) = 199.00, \quad EV(R) = 201.95.$$

EV ranking: $P > R > Q$.

Step 3. Maximin criterion.

$$\min(P) = 65, \quad \min(Q) = 10, \quad \min(R) = 40.$$

Maximin ranking: $P > R > Q$.

Step 4. Minimax Regret criterion. State best payoffs: $\{320, 250, 130, 65\}$.

Alternative	HS	HC	LS	LC	Max regret
P	10	5	0	0	10
Q	0	0	50	55	55
R	5	0	25	25	25

Minimax Regret ranking (lower is better): $P < R < Q$.

Step 5. Staged decision tree and final choice. Filtering rule: keep best 2 EV options, then choose smaller Minimax Regret.

Decision tree
D_0: $\{P, Q, R\}$
$\rightarrow D_1$ (EV filter): keep $\{P, R\}$
$\rightarrow D_2$ (Minimax Regret): compare $P(10)$ vs $R(25)$
Leaves: P, R
Final leaf selected: P

Final explanation: P stays strongest across expected value, worst-case floor, and regret, so the staged process ends at P.

2.

Problem description

An online retailer must choose one pricing strategy among five options S_1, S_2, S_3, S_4, S_5. Demand states are optimistic (0.45), regular (0.35), and weak (0.20). Payoffs (thousand USD):

State	Probability	S_1	S_2	S_3	S_4	S_5
Optimistic	0.45	150	140	170	130	160
Regular	0.35	90	100	80	110	95
Weak	0.20	20	45	-10	60	35

Apply EV, Maximin, and Minimax Regret step by step, generate full ordered rankings, and then apply this staged rule: keep best 4 EV options → keep best 3 Maximin options → choose smallest Minimax Regret.

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Step 1. EV calculations.

$$EV(S_1) = 103.00,$$
$$EV(S_2) = 107.00,$$
$$EV(S_3) = 102.50,$$
$$EV(S_4) = 109.00,$$
$$EV(S_5) = 112.25.$$

EV ranking:

$$S_5\,(112.25) > S_4\,(109.00) > S_2\,(107.00) > S_1\,(103.00) > S_3\,(102.50).$$

Step 2. Maximin calculations.

$$\min(S_1) = 20,\ \min(S_2) = 45,\ \min(S_3) = -10,\ \min(S_4) = 60,\ \min(S_5) = 35.$$

Maximin ranking:

$$S_4\,(60) > S_2\,(45) > S_5\,(35) > S_1\,(20) > S_3\,(-10).$$

Step 3. Minimax Regret calculations. Best state payoffs:

$$\max(\text{Optimistic}) = 170, \quad \max(\text{Regular}) = 110, \quad \max(\text{Weak}) = 60.$$

Regret table:

Alternative	Optimistic	Regular	Weak	Max regret
S_1	20	20	40	40
S_2	30	10	15	30
S_3	0	30	70	70
S_4	40	0	0	40
S_5	10	15	25	25

Minimax Regret ranking (lower is better):

$$S_5\,(25) > S_2\,(30) > S_1\,(40) = S_4\,(40) > S_3\,(70).$$

Step 4. Decision tree for the staged rule.

- EV top 4: $\{S_5, S_4, S_2, S_1\}$.
- On this subset, Maximin top 3: $S_4(60), S_2(45), S_5(35)$, so keep $\{S_4, S_2, S_5\}$.
- On this subset, Minimax Regret values are $S_4(40), S_2(30), S_5(25)$.

Decision tree
D_0: $\{S_1, S_2, S_3, S_4, S_5\}$
$\rightarrow D_1$ (EV filter): $\{S_5, S_4, S_2, S_1\}$
$\rightarrow D_2$ (Maximin filter): $\{S_4, S_2, S_5\}$
$\rightarrow D_3$ (Minimax Regret choice): compare $S_4(40), S_2(30), S_5(25)$
Leaves: S_4, S_2, S_5
Final leaf selected: S_5

Step 5. Final conclusion. The final decision is S_5 because it leads EV and has the smallest maximum regret among the finalists.